HOMEWORK 2: Function Approximation MLP

Import Libraries

In [207]:
```
from keras.utils import to_categorical
from keras.datasets import fashion_mnist
from keras.models import Model
from keras.layers import Input, Conv2D, MaxPooling2D, Flatten, Dens
e, Dropout
from keras.optimizers import SGD
from keras.callbacks import ModelCheckpoint, EarlyStopping, LambdaC
allback
import keras.backend as K
import datetime
import cv2

import os
import sys
import time
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from scipy.stats import multivariate_normal
from IPython import display
display.clear_output()
```

In [2]:
```
from tensorflow.python.client import device_lib
print(device_lib.list_local_devices())
```

```
[name: "/device:CPU:0"
device_type: "CPU"
memory_limit: 268435456
locality {
}
incarnation: 6959318980692923618
, name: "/device:GPU:0"
device_type: "GPU"
memory_limit: 6667611669
locality {
  bus_id: 1
  links {
  }
}
incarnation: 5314801303220798765
physical_device_desc: "device: 0, name: GeForce GTX 1080, pci bus
id: 0000:01:00.0, compute capability: 6.1"
]
```

Some Functions

In [63]:
```
def generate_data(n,  r1= (-2,2), r2= (-4,4), m= np.array([0.0,0.
0]), c= np.array([[0.3,-0.5],[-0.5,2]]), export_path="./data/train.s
cv", verbose = 1):

    # generate data
    x = np.random.rand(n, 2)
    x[:, 0] = x[:, 0] * (r1[1] - r1[0]) + r1[0]
    x[:, 1] = x[:, 1] * (r2[1] - r2[0]) + r2[0]

    # calculate gaussian
    y = multivariate_normal.pdf(x, mean = m, cov=c)

    # debug
    if verbose == 1:
        print("x1: ", np.min(x[:, 0]), np.max(x[:, 0]))
        print("x2: ", np.min(x[:, 1]), np.max(x[:, 1]))
        print("x shape: ", x.shape)
        print("y: ", np.min(y), np.max(y))
        print("y shape: ", y.shape)
    # if

    # save data
    db_table = dict(
        data = {"x1": x[:, 0], "x2": x[:, 1], "y": y},
        columns = ["x1", "x2", "y"]
    )

    ds_data = pd.DataFrame(**db_table)

    save_dir = os.path.dirname(export_path)
    if save_dir != "" and os.path.exists(save_dir) == False: os.make
dirs(save_dir)
    ds_data.to_csv(export_path, index=False)

    return ds_data
# generate_dataset

def load_data(csv_file):
    df_data  = pd.read_csv(csv_file)
    return df_data[["x1", "x2"]].values, df_data[["y"]].values.flatt
en()
# load_data

def load_dataset(csv_train = "./data/train.csv",
                 csv_valid = "./data/valid.csv",
                 csv_test  = "./data/test.csv"):
    # test
    x_test, y_test = load_data(csv_test)

    #train
    x_train, y_train = load_data(csv_train)

    #valid
    x_valid, y_valid = load_data(csv_valid)
```

```
    return x_train, y_train, x_valid, y_valid, x_test, y_test
# load_dataset

def visualize_dataset(x, y, color = "green", title = "Training Data
\n", ax = None):
    if ax is None:
        plt.figure(figsize=(16, 6))
        ax = plt.subplot(1,2,1, projection='3d')
    # if

    surf = ax.scatter(x[:,0], x[:,1], y, c = color)
    ax.set_frame_on
    ax.set_xlabel('x1', labelpad = 5, fontdict=dict(fontsize = 14))
    ax.set_ylabel('x2', labelpad = 5, fontdict=dict(fontsize = 14))
    ax.set_zlabel("y", labelpad = 5, fontdict=dict(fontsize = 14))
    plt.legend(['Gaussian values'])
    plt.title(title, fontdict=dict(fontsize = 20))
# visualize_dataset

def normalize_data(x_train, x_valid, x_test):
    x_train, y_train, x_valid, y_valid, x_test, y_test = load_datase
t(csv_train="./data/train.csv",

csv_valid="./data/valid.csv",

csv_test="./data/test.csv")
    v_mean, v_std = None, None

    # Visualize data
    print("********************")
    print("Befor Normalization")
    print("********************")
    print("Train min=%.3f, max=%.3f" % (x_train.min(), x_train.max
()))
    print("Valid min=%.3f, max=%.3f" % (x_valid.min(), x_valid.max
()))
    print("Test min=%.3f, max=%.3f" % (x_test.min(), x_test.max()))

    # Visualize data
    if v_mean is None and v_std is None: # Prevent normalization wit
h many times
        v_mean = np.mean(x_train, axis = 0)
        v_std  = np.std(x_train, axis = 0)

        x_train = (x_train - v_mean) / v_std
        x_valid = (x_valid - v_mean) / v_std
        x_test  = (x_test - v_mean) / v_std
    # if
    print("\n\n")
    print("Mean = %s, Std = %s"%(str(v_mean), str(v_std)))

    print("********************")
    print("After Normalization")
    print("********************")
    print("Train min=%.3f, max=%.3f" % (x_train.min(), x_train.max
()))
    print("Valid min=%.3f, max=%.3f" % (x_valid.min(), x_valid.max
()))
    print("Test min=%.3f, max=%.3f" % (x_test.min(), x_test.max()))
```

```
    return x_train, x_valid, x_test
# build_data

def summarize_diagnostics(history):
    # plot loss
    plt.figure(figsize=(10,10))
    plt.subplot(211)
    plt.title("Mean Squared Error")
    plt.plot(history.history["loss"], color="blue", label="train")
    plt.plot(history.history['val_loss'], color="orange", label='tes
t')
# summarize_diagnostics

def visualize_logs(visualize_data, group_data, figsize = (10, 5), is
_show = True, label_size = 12, title_size = 16, save_path = None):
    """
    visualize_data = df_logs[50:].to_dict(orient='list')
    group_data     = [{"x": "epoch", "y": ["mean_squared_error", "va
l_mean_squared_error"],
                       "title": "Error at Epoch {epoch}: {val_mean_s
quared_error:.4f}\n",
                       "style": "min"},
                      {"x": "epoch", "y": ["loss", "val_loss"],
                       "title": "Loss at Epoch {epoch}: {val_loss:.4
f}\n",
                       "style": "min"}]
    figsize        = (10, 5)
    visualize_logs(visualize_data, group_data, figsize)
    """
    import matplotlib.ticker as ticker
    import matplotlib.pyplot as plt, numpy as np

    fig = plt.figure(figsize=figsize)
    for idx, group in enumerate(group_data):
        x_axis_name, y_axis_names, title, style = group["x"], group
["y"], group["title"], group["style"]
        x_values = visualize_data[x_axis_name]
        end_logs = dict([(x_axis_name, x_values[-1])])

        ax = plt.subplot(1, len(group_data), idx%len(group_data) +
1)
        ax.xaxis.set_major_locator(ticker.MaxNLocator(integer=True))

        for y_axis_name in y_axis_names:
            y_values = visualize_data[y_axis_name]
            if style is not None and style != "":
                y_values = []
                for i in range(len(visualize_data[y_axis_name])):
                    if style == "median":
                        y_values.append(np.median(visualize_data[y_a
xis_name][:i+1]))
                    elif style == "avg":
                        y_values.append(np.average(visualize_data[y_
axis_name][:i+1]))
                    elif style == "max":
                        y_values.append(np.max(visualize_data[y_axis
_name][:i+1]))
                    elif style == "min":
                        y_values.append(np.min(visualize_data[y_axis
_name][:i+1]))
```

```
            # if
            end_logs.update(dict([(y_axis_name, y_values[-1])]))
            ax.plot(x_values, y_values)
        pass
        title = title.format(**end_logs)

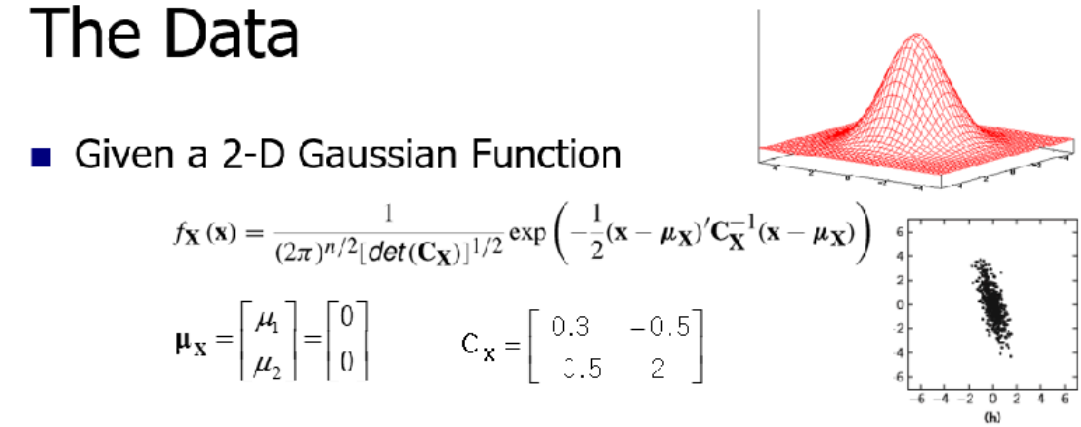
        ax.set_xlabel(x_axis_name, fontsize = label_size)
        ax.legend(y_axis_names, loc='upper left', fontsize = label_s
ize)
        ax.set_title(title, fontsize = title_size)
    # for

    if save_path is not None:
        save_dir = os.path.dirname(save_path)
        if save_dir != "" and os.path.exists(save_dir) == False: os.
makedirs(save_dir)
        fig.savefig(save_path)
    # if

    if is_show == True:
        plt.show()
        return None
    return fig
# visualize_logs
```

1. DATASET

In [213]:
```
image = plt.imread("img/problem.png")
plt.figure(figsize=(14,14))
plt.axis("off"), plt.imshow(image);
```



1.1. Generate data

In [64]:
```
rangex1 = (-2,2)
rangex2 = (-4,4)
mean = np.array([0.0,0.0])
cov = np.array([[0.3,-0.5],[-0.5,2]])

generate_data(n=20000, r1=rangex1, r2=rangex2, m=mean, c=cov, export
_path="./data/train.csv", verbose = 1);
generate_data(n=100, r1=rangex1, r2=rangex2, m=mean, c=cov, export_p
ath="./data/test.csv", verbose = 1);
generate_data(n=100, r1=rangex1, r2=rangex2, m=mean, c=cov, export_p
ath="./data/valid.csv", verbose = 1);
```

```
x1:  -1.9999525035989705 1.9999863989902917
x2:  -3.9997789127804655 3.999758427455422
x shape:  (20000, 2)
y:  5.326482001244772e-14 0.2689785062092115
y shape:  (20000,)
x1:  -1.9695441217668828 1.936559384904803
x2:  -3.915986121969407 3.9974477378761977
x shape:  (100, 2)
y:  7.641211152445757e-12 0.2504619331951585
y shape:  (100,)
x1:  -1.979927522362798 1.9647878568160961
x2:  -3.995542572328751 3.9577867133235163
x shape:  (100, 2)
y:  1.7406205146015622e-12 0.2631298278954595
y shape:  (100,)
```

1.2. Load dataset

In [65]:
```
x_train, y_train, x_valid, y_valid, x_test, y_test = load_dataset(cs
v_train="./data/train.csv", csv_valid="./data/valid.csv", csv_test
="./data/test.csv")
```

1.3. Visualize dataset

In [66]:
```
print("Training samples: ", len(x_train), " -- Shape: ", x_train.shape, y_train.shape)
print("Validating samples: ", len(x_valid), " -- Shape: ", x_valid.shape, y_valid.shape)
print("Testing samples: ", len(x_test), " -- Shape: ", x_test.shape, y_test.shape)

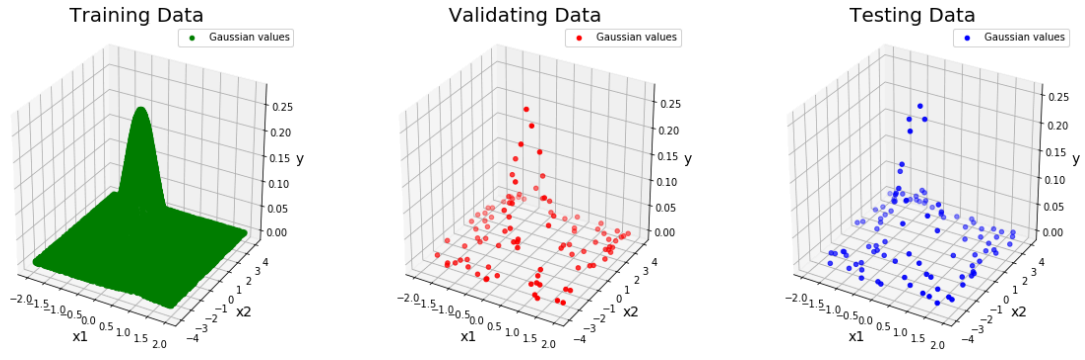
plt.figure(figsize=(20, 6))

ax1 = plt.subplot(1,3,1, projection='3d')
visualize_dataset(x_train, y_train, color = "green", title = "Training Data\n", ax = ax1)

ax2 = plt.subplot(1,3,2, projection='3d')
visualize_dataset(x_valid, y_valid, color = "red", title = "Validating Data\n", ax = ax2)

ax3 = plt.subplot(1,3,3, projection='3d')
visualize_dataset(x_test, y_test, color = "blue", title = "Testing Data\n", ax = ax3)
```

```
Training samples:  20000  -- Shape:  (20000, 2) (20000,)
Validating samples:  100  -- Shape:  (100, 2) (100,)
Testing samples:  100  -- Shape:  (100, 2) (100,)
```



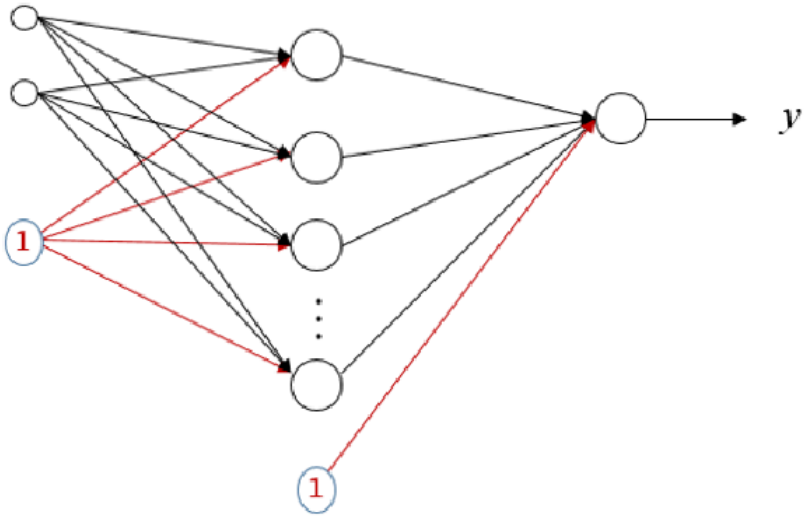
1.4. Normalize Data

In [67]:
```
# x_train, x_valid, x_test = normalize_data(x_train, x_valid, x_test)
```

2. NETWORK STRUCTURE

In [217]:
```
image = plt.imread("img/model.png")
plt.figure(figsize=(14,14))
plt.axis("off"), plt.imshow(image);
```

- MLP with one hidden layer
  - 2 input nodes (+ bias)
  - 1 output node
  - H hidden nodes



In [191]:
```
def build_model(H=10):
    inp = Input(shape=(2,), name='Input')
    x = Dense(units=H, activation='sigmoid', name='fc')(inp)
    output_label = Dense(units=1, activation='sigmoid', name='gauss
_apxm')(x)

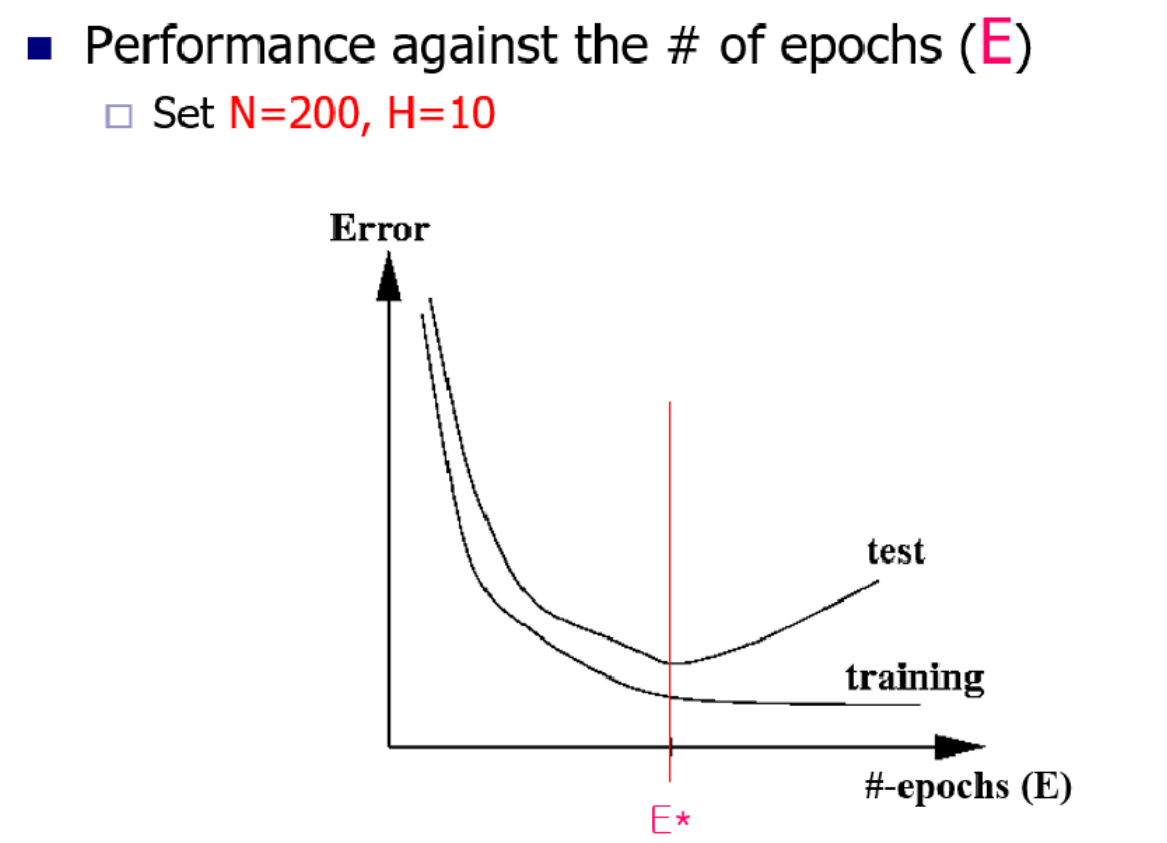
    model = Model(inputs=inp, outputs=output_label)
    model.summary()
    return model
#model
```

In [193]:
```
# build a sample model with hidden node = 10
model = build_model(H=10)
```

```
Model: "model_5"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```

3. EXPERIMENT 1:

In [214]:
```
image = plt.imread("img/exp1.png")
plt.figure(figsize=(14,14))
plt.axis("off"), plt.imshow(image);
```



3.1. Train Data

In [73]:
```
x_exp1_train, y_exp1_train = x_train[:200], y_train[:200]
x_exp1_valid, y_exp1_valid = x_valid, y_valid

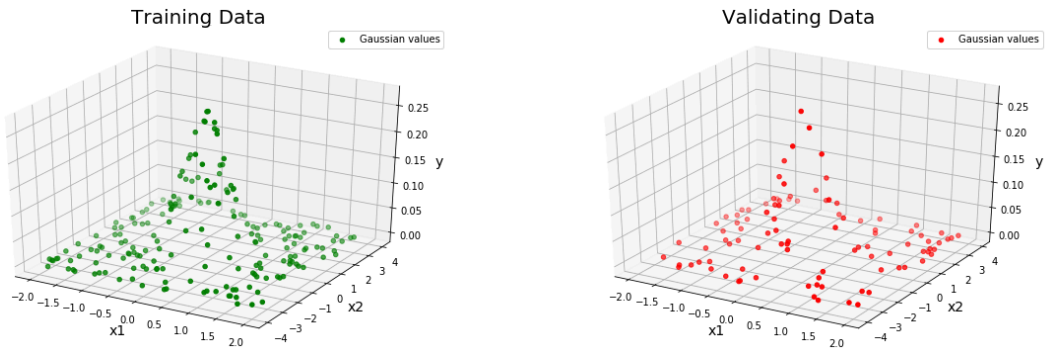
print(f"Train: x-shape = {x_exp1_train.shape} y-shape = {y_exp1_train.shape}")
print(f"Valid: x-shape = {x_exp1_valid.shape} y-shape = {y_exp1_valid.shape}")

plt.figure(figsize=(20, 6))

ax1 = plt.subplot(1,2,1, projection='3d')
visualize_dataset(x_exp1_train, y_exp1_train, color = "green", title = "Training Data\n", ax = ax1)

ax2 = plt.subplot(1,2,2, projection='3d')
visualize_dataset(x_exp1_valid, y_exp1_valid, color = "red", title = "Validating Data\n", ax = ax2)
```

```
Train: x-shape = (200, 2) y-shape = (200,)
Valid: x-shape = (100, 2) y-shape = (100,)
```



3.2. Model

In [190]:
```
exp1_model = build_model(H=10)
```

```
Model: "model_3"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```

3.3. Training

In [147]:
```
K.clear_session()
exp1_model = build_model(H=10)

# OPTIMIZATION
exp1_model.compile(loss='mean_squared_error', optimizer=SGD(lr=0.0
1), metrics=['mean_squared_error'])

# CALLBACKS
callbacks = []

# + ModelCheckPoint
save_dir = "./data/model/train_01"
template_name= "funcApxm_model_01_{epoch:02d}_{val_loss:.3f}_val_ac
c-{val_mean_squared_error:.3f}.h5"
if os.path.exists(save_dir) == False: os.makedirs(save_dir)
modekcheckpoint_callback = ModelCheckpoint(filepath = os.path.join
(save_dir, template_name),
                                           monitor='val_loss',
                                           verbose=0,
                                           save_best_only=True,
                                           save_weights_only=False,
# only weight,
                                           mode='auto', # min, max,
auto
                                           period=1   # bao nhieu ep
och moi xem xet 1 lan
                                          )
callbacks.append(modekcheckpoint_callback)

# + Early Stopping
earlystopping_callback = EarlyStopping(monitor='val_loss', patience
= 10)
callbacks.append(earlystopping_callback)


# + Visualize logs
review_logs = {}
def custom_on_epoch_end(self, epoch, logs, **kwargs):
    review_logs = kwargs["review_logs"]
    if epoch == 0:
        review_logs["epoch"] = [epoch]
        for key in logs.keys(): review_logs[key] = [logs[key]]
    # if
    review_logs["epoch"].append(epoch)
    for key in logs.keys(): review_logs[key].append(logs[key])

    visualize_data = review_logs
    group_view     = [{"x": "epoch", "y": ["loss", "val_loss"],
                       "title": "Loss at Epoch {epoch}: {val_loss:.
6f}\n",
                       "style": "median"},
                      {"x": "epoch", "y": ["mean_squared_error", "v
al_mean_squared_error"],
                       "title": "MSE at Epoch {epoch}: {val_mean_sq
uared_error:.6f}\n",
```

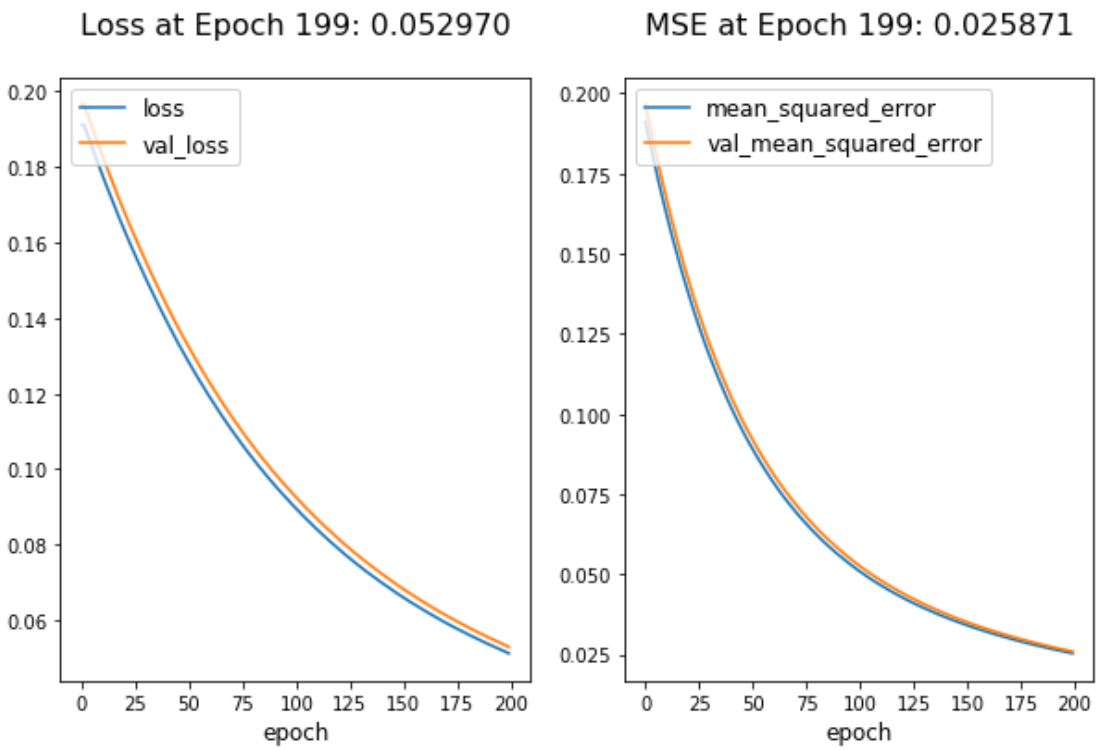
```
                        "style": None}
                    ]
    fig = visualize_logs(visualize_data, group_view, figsize = (10,
6), is_show = False, save_path = os.path.join("./data/model/train_0
1", "logs.png"))
    plt.close()
    if epoch == 0:
        display.display(fig, display_id="logs")
    else:
        display.update_display(fig, display_id="logs")
    pass
# custom_on_epoch_end

custom_callbacks = LambdaCallback()
custom_callbacks.on_epoch_end = lambda epoch, logs: custom_on_epoch
_end(custom_callbacks, epoch, logs, review_logs = review_logs)
callbacks.append(custom_callbacks)

# TRAINING
batch_size = 128
epochs = 200

starting_time = time.time()
history = exp1_model.fit(x_exp1_train, y_exp1_train,
                    validation_data=(x_exp1_valid, y_exp1_valid),
                    batch_size=batch_size,
                    epochs=epochs,
                    callbacks = callbacks,
                    verbose = 0)
print('> training time is %.4f minutes' % ((time.time() - starting_
time)/60))
```

```
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```




```
> training time is 1.1652 minutes
```

3.4. Evaluating

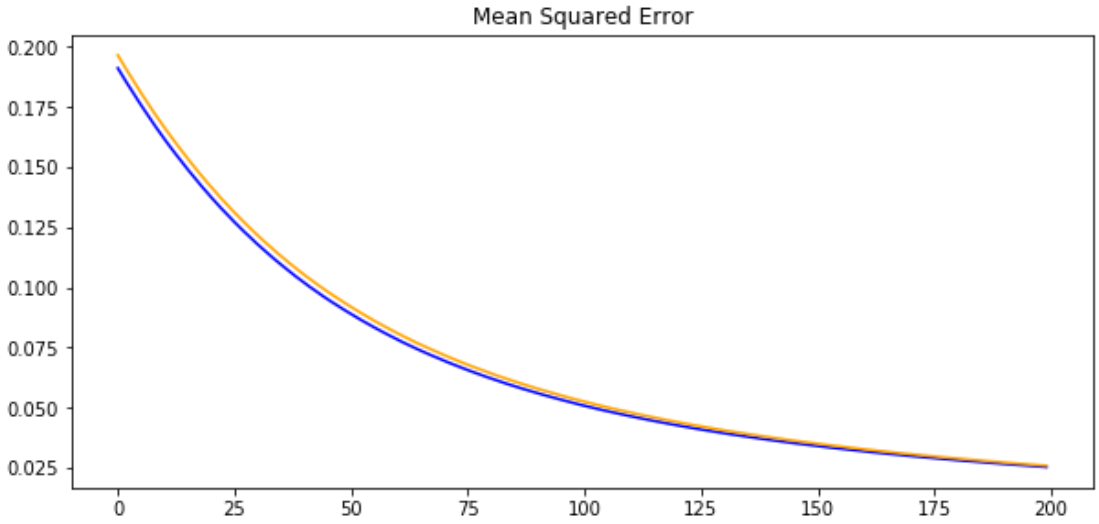
In [148]:
```
epoch_opt = np.argmin(history.history["val_mean_squared_error"])
```

In [149]:
```
save_name = template_name.format(epoch = epoch_opt + 1,
                  val_mean_squared_error = history.history["val_mean
_squared_error"][epoch_opt],
                  val_loss= history.history["val_loss"][epoch_opt])
save_path = os.path.join(save_dir, save_name)
save_path
```

Out[149]:
```
'./data/model/train_01\\funcApxm_model_01_200_0.026_val_acc-0.026.
h5'
```

In [150]:
```
from keras.models import load_model
model = load_model(save_path)
score = model.evaluate(x_test, y_test, verbose=0)
print('Test loss:', score[0])
print('Test accuracy:', score[1])
summarize_diagnostics(history)
```

```
Test loss: 0.026295990273356436
Test accuracy: 0.026295991614460945
```

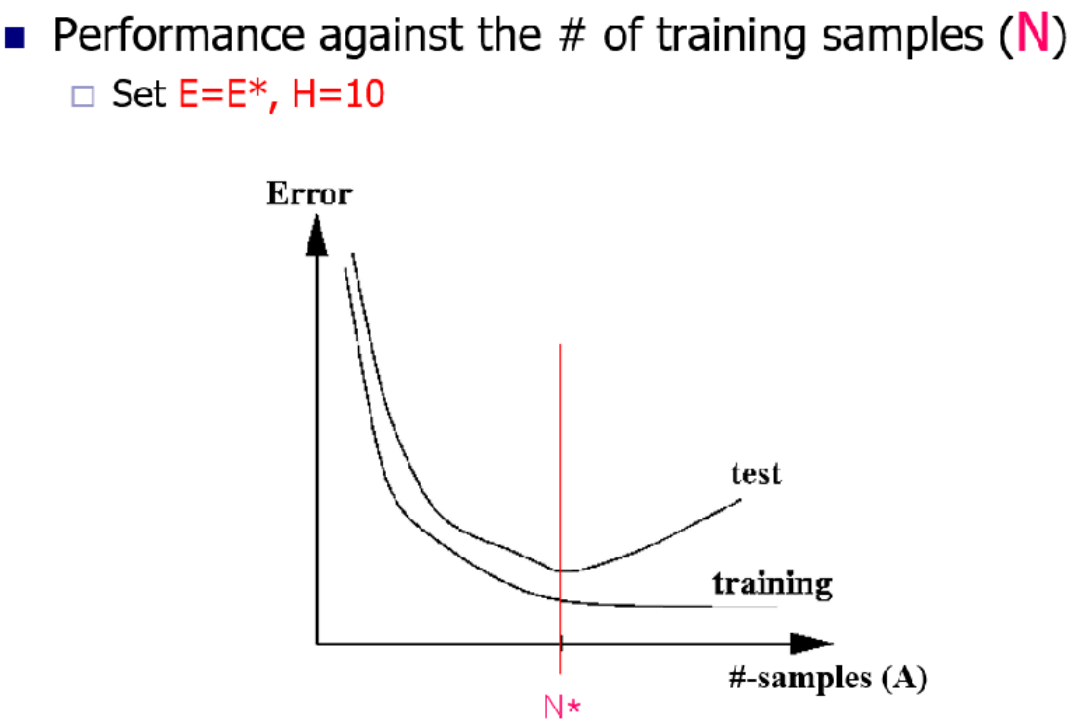


3.5. Discussion

- Training with HiddenNode = 10, Maximum Epochs = 200, EarlyStopping = 10
- No overfit and achieve optimization error at epoch = 200 with accuracy = 0.026295991614460945

4. EXPERIMENT 2:

In [215]:
```
image = plt.imread("img/exp2.png")
plt.figure(figsize=(14,14))
plt.axis("off"), plt.imshow(image);
```



- Setup:
  - E* = 200
  - H = 10
  - N = 0 --> 2000, step = 200

4.1. Train Data

In [151]:
```
x_exp2_train, y_exp2_train = [], []
x_exp2_valid, y_exp2_valid = x_valid, y_valid
print("Train: ")
for size in range(200, 2001, 200):
    x_exp2_train.append(x_train[:size])
    y_exp2_train.append(y_train[:size])
    print(f"+ Size: {size} {x_exp2_train[-1].shape} {y_exp2_train[-1].shape}")
# for
print(f"Valid: {x_exp2_valid.shape} {y_exp2_valid.shape}")
```

```
Train:
+ Size: 200 (200, 2) (200,)
+ Size: 400 (400, 2) (400,)
+ Size: 600 (600, 2) (600,)
+ Size: 800 (800, 2) (800,)
+ Size: 1000 (1000, 2) (1000,)
+ Size: 1200 (1200, 2) (1200,)
+ Size: 1400 (1400, 2) (1400,)
+ Size: 1600 (1600, 2) (1600,)
+ Size: 1800 (1800, 2) (1800,)
+ Size: 2000 (2000, 2) (2000,)
Valid: (100, 2) (100,)
```

4.2. Build model

In [152]:
```
exp2_model = build_model(H = 10)
```

```
Model: "model_2"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```

4.3. Training

In [153]:

```
def train_model(train_id, x_local_train, y_local_train,
                x_local_valid, y_local_valid, nb_epochs = 1000):

    K.clear_session()
    exp2_model = build_model(H=10)

    # OPTIMIZATION
    exp2_model.compile(loss='mean_squared_error', optimizer=SGD(lr=
0.01), metrics=['mean_squared_error'])

    # CALLBACKS
    callbacks = []

    # + ModelCheckPoint
    save_dir = f"./data/model/train_02/{train_id}"
    if os.path.exists(save_dir)==False: os.makedirs(save_dir)
    template_name= "funcApxm_model_01_{epoch:02d}_{val_loss:.3f}_va
l_acc-{val_mean_squared_error:.3f}.h5"
    if os.path.exists(save_dir) == False: os.makedirs(save_dir)
    modekcheckpoint_callback = ModelCheckpoint(filepath = os.path.j
oin(save_dir, template_name),
                                               monitor='val_loss',
                                               verbose=0,
                                               save_best_only=True,
                                               save_weights_only=Fa
lse, # only weight,
                                               mode='auto', # min,
max, auto
                                               period=1  # bao nhie
u epoch moi xem xet 1 lan
                                              )
    callbacks.append(modekcheckpoint_callback)

    # + Early Stopping
    # earlystopping_callback = EarlyStopping(monitor='val_loss', pa
tience= 10)
    # callbacks.append(earlystopping_callback)

    # + Visualize logs
    review_logs = {}
    def custom_on_epoch_end(self, epoch, logs, **kwargs):
        review_logs = kwargs["review_logs"]
        if epoch == 0:
            review_logs["epoch"] = [epoch]
            for key in logs.keys(): review_logs[key] = [logs[key]]
        # if
        review_logs["epoch"].append(epoch)
        for key in logs.keys(): review_logs[key].append(logs[key])

        visualize_data = review_logs
        group_view     = [{"x": "epoch", "y": ["loss", "val_loss"],
                           "title": "Loss at Epoch {epoch}: {val_lo
ss:.6f}\n",
                           "style": "median"},
                          {"x": "epoch", "y": ["mean_squared_erro
```

```
r", "val_mean_squared_error"],
                                "title": "MSE at Epoch {epoch}: {val_mea
n_squared_error:.6f}\n",
                                "style": None}
                               ]
        fig = visualize_logs(visualize_data, group_view, figsize =
(10, 6), is_show = False,
                                 save_path = os.path.join(f"./data/mode
l/train_02/{train_id}", "logs.png"))
        plt.close()
        if epoch == 0:
            display.display(fig, display_id=f"logs_{train_id}")
        else:
            display.update_display(fig, display_id=f"logs_{train_i
d}")
        pass
    # custom_on_epoch_end

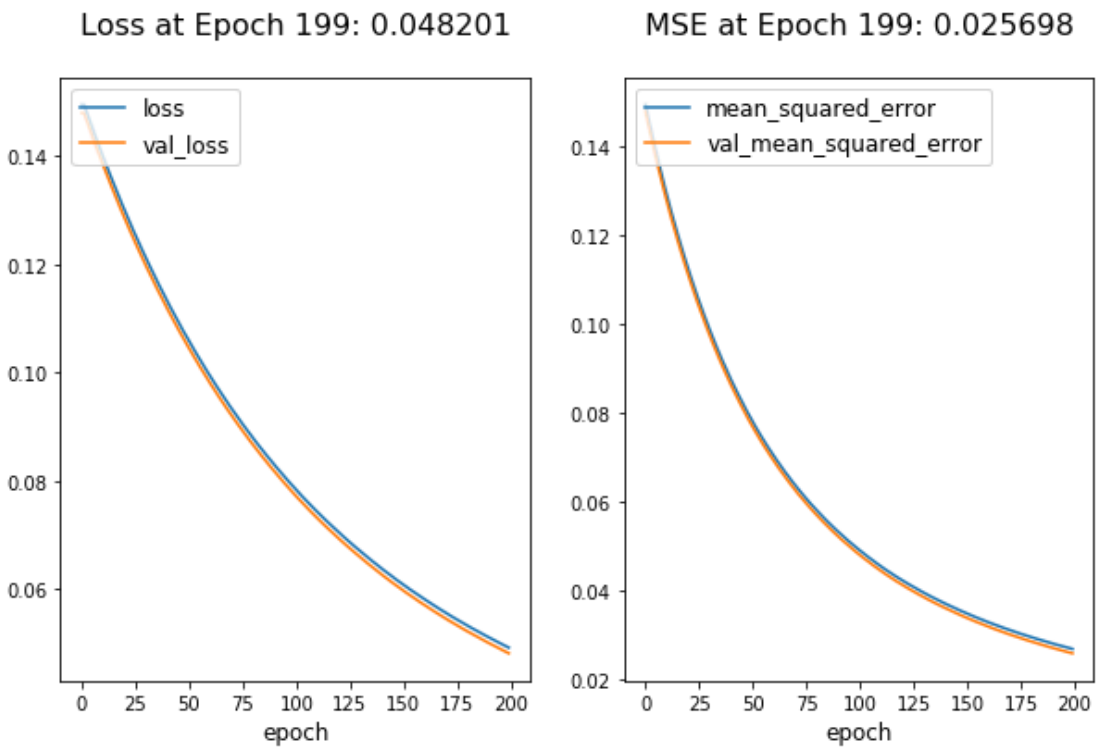
    custom_callbacks = LambdaCallback()
    custom_callbacks.on_epoch_end = lambda epoch, logs: custom_on_e
poch_end(custom_callbacks, epoch, logs, review_logs = review_logs)
    callbacks.append(custom_callbacks)

    # TRAINING
    batch_size = 128
    epochs = nb_epochs

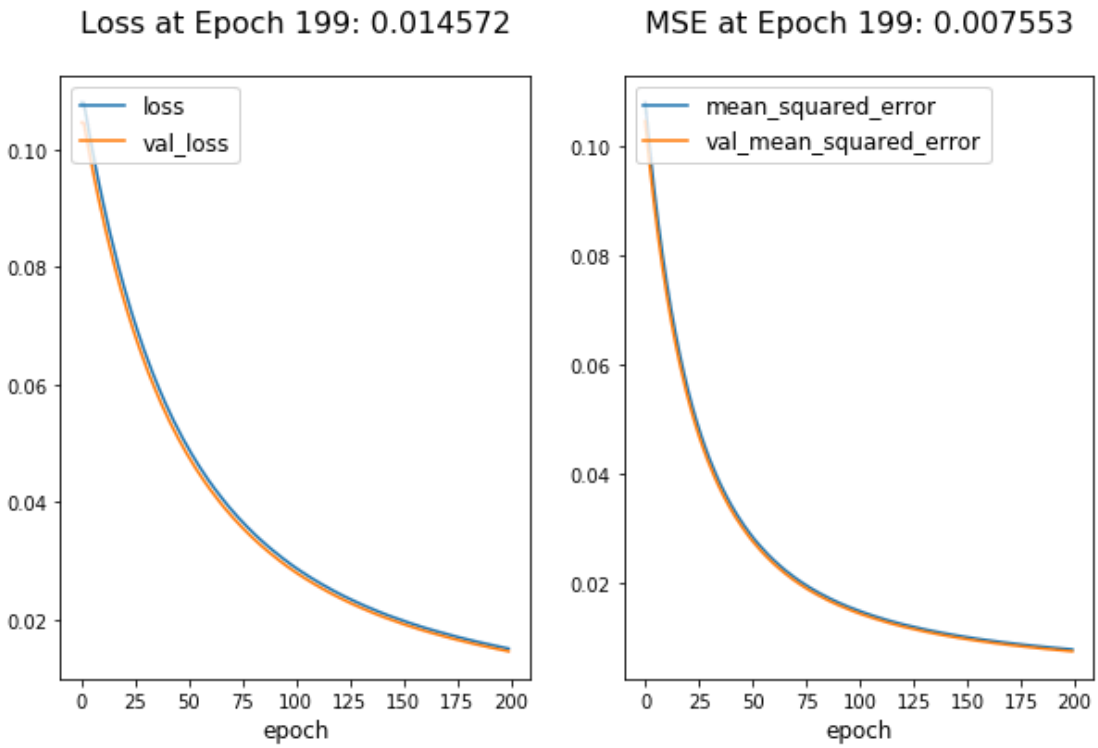
    starting_time = time.time()
    history = exp2_model.fit(x_local_train, y_local_train,
                         validation_data=(x_local_valid, y_local_val
id),
                         batch_size=batch_size,
                         epochs=epochs,
                         callbacks = callbacks,
                         verbose = 0)
    print('> training time is %.4f minutes' % ((time.time() - start
ing_time)/60))
    return history
# train_model
```

In [154]:
```
history = []
for i in range(len(x_exp2_train)):
    print(f"Train id: {i}")
    temp_history = train_model(i, x_exp2_train[i], y_exp2_train[i],
x_exp2_valid, y_exp2_valid, nb_epochs = 200)
    history.append(temp_history)
# for
```

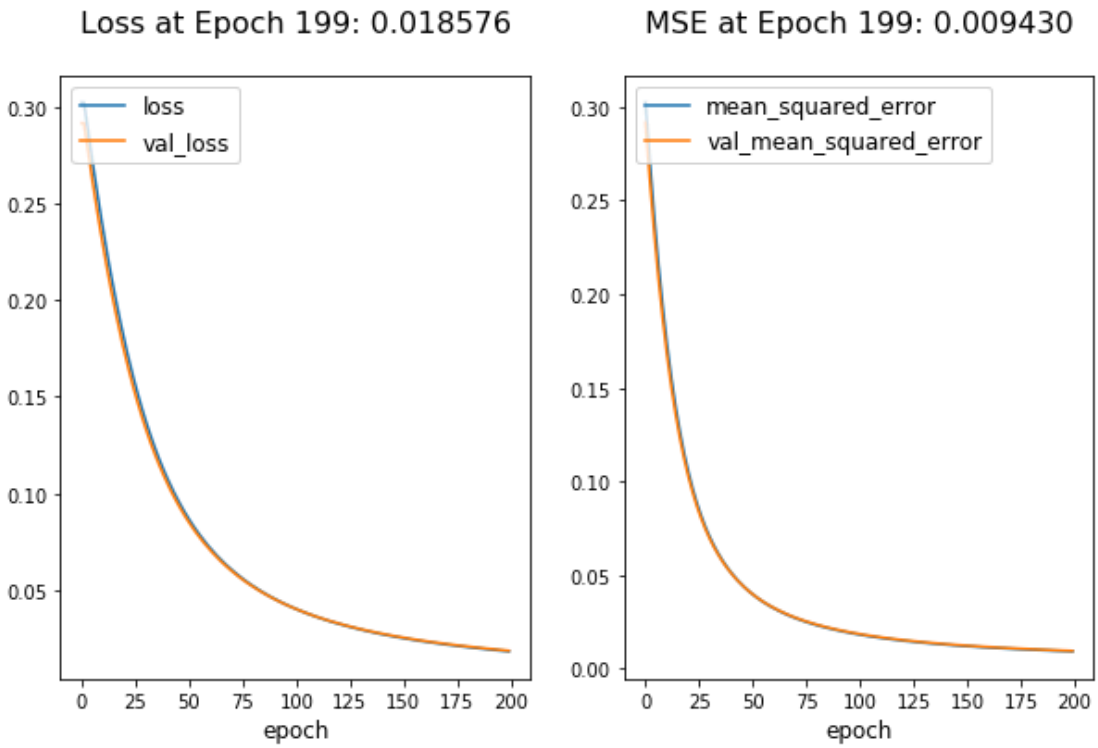
```
Train id: 0
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```



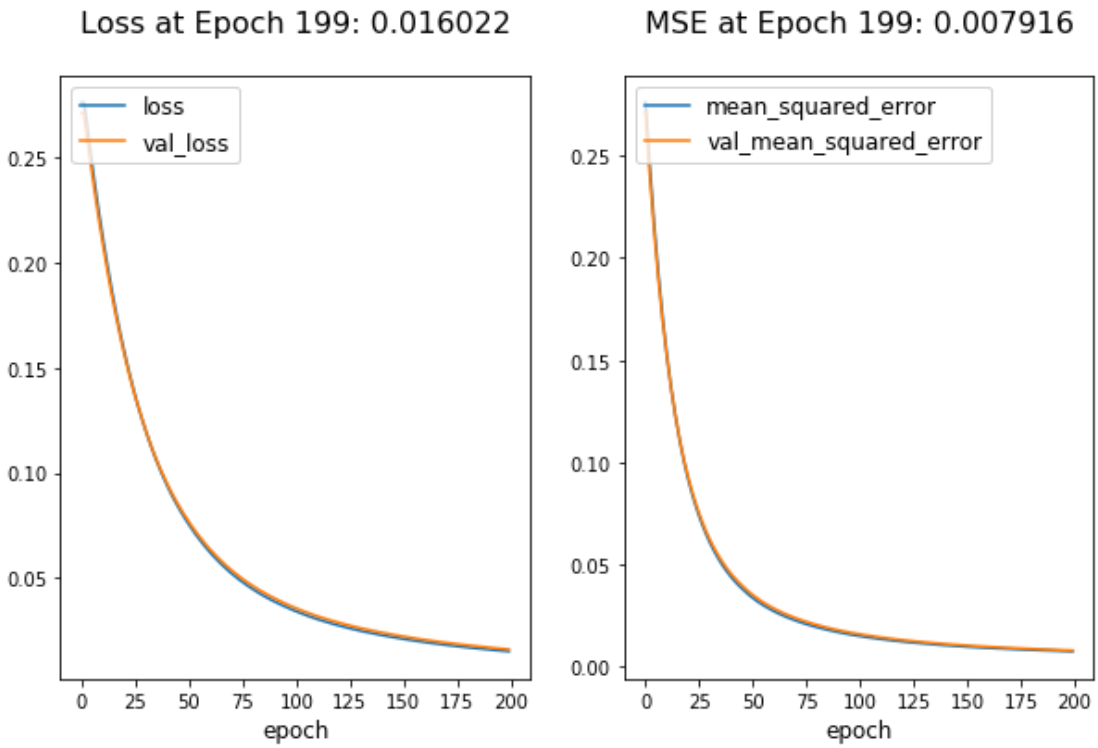

```
> training time is 1.2596 minutes
Train id: 1
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```



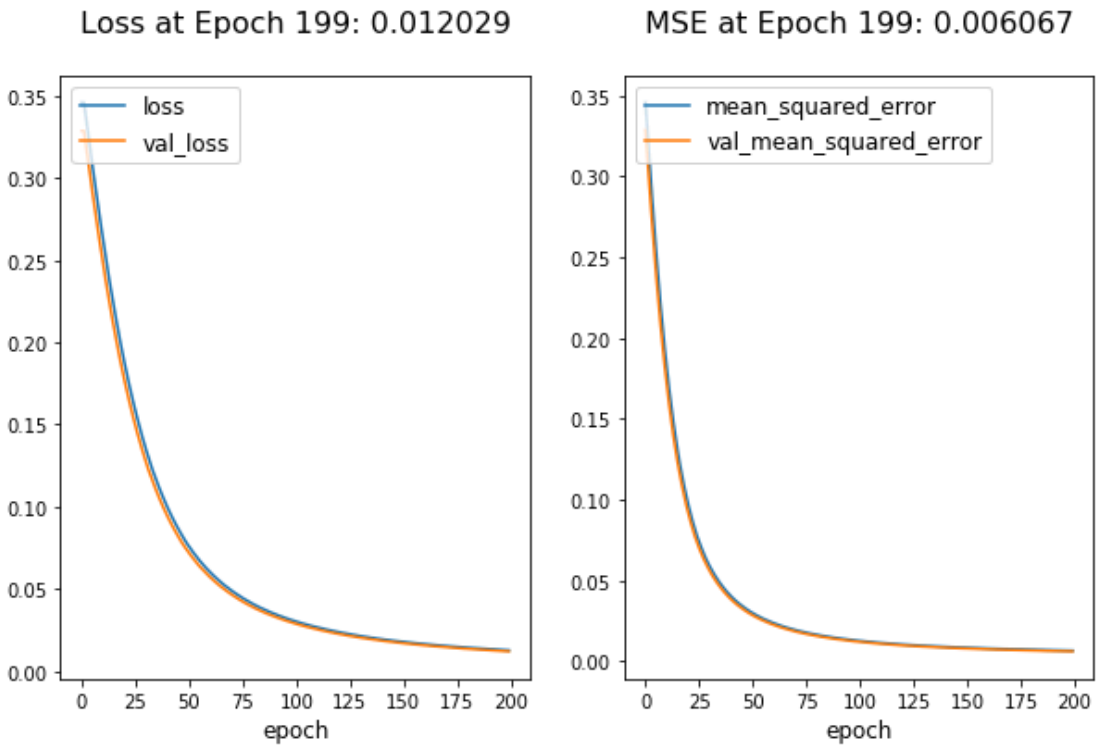

```
> training time is 1.3262 minutes
Train id: 2
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```



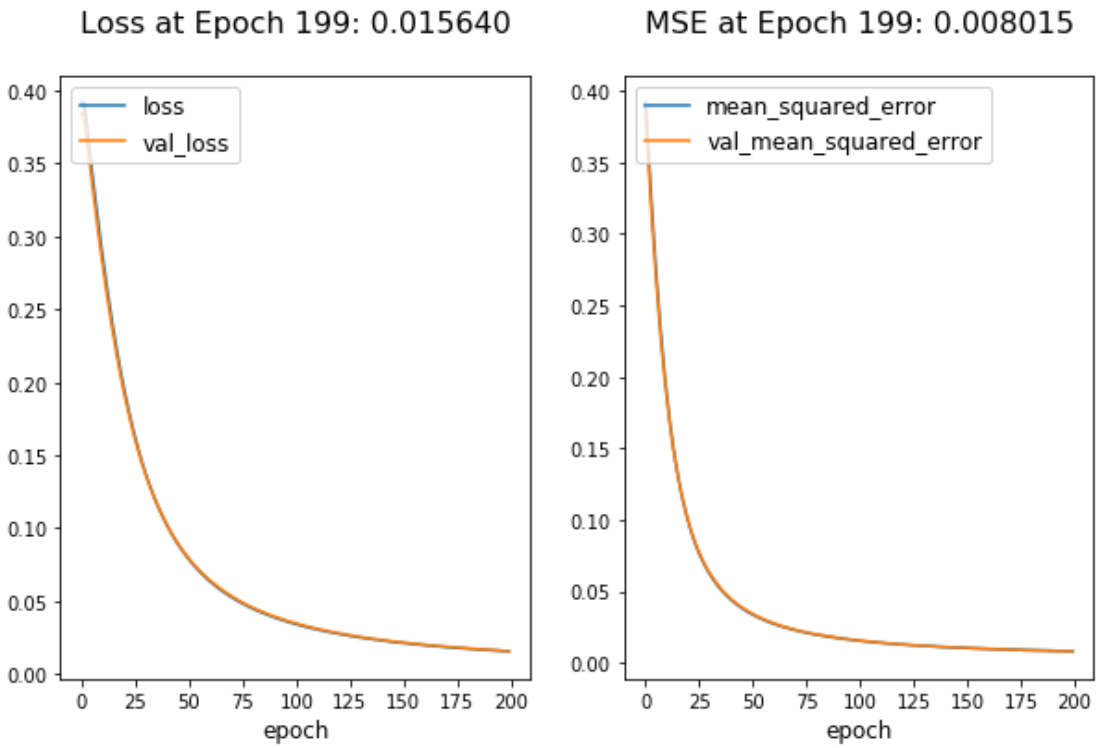

```
> training time is 1.3511 minutes
Train id: 3
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```



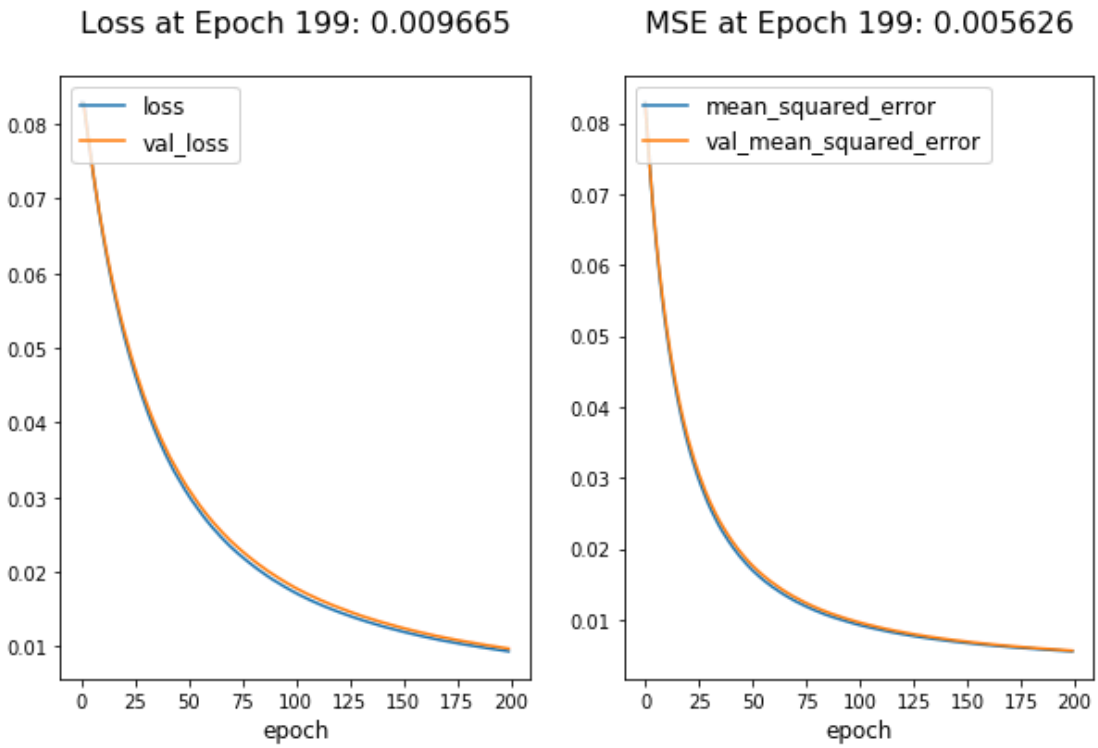

```
> training time is 1.3777 minutes
Train id: 4
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```



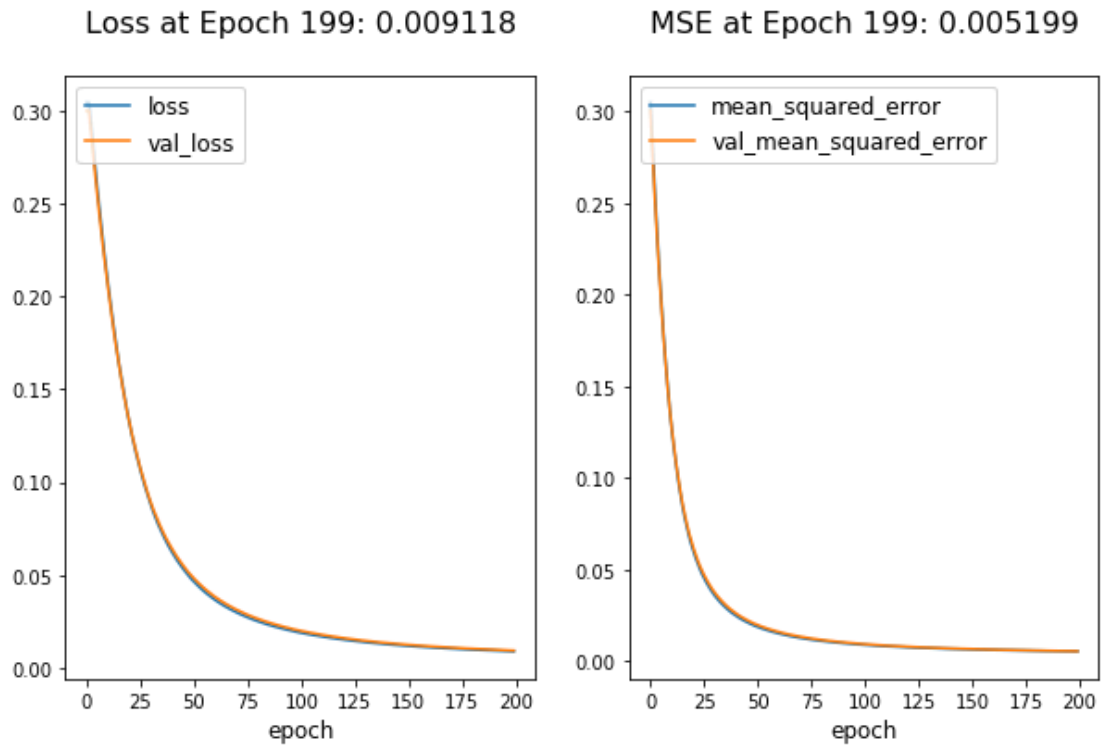

```
> training time is 1.4272 minutes
Train id: 5
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```



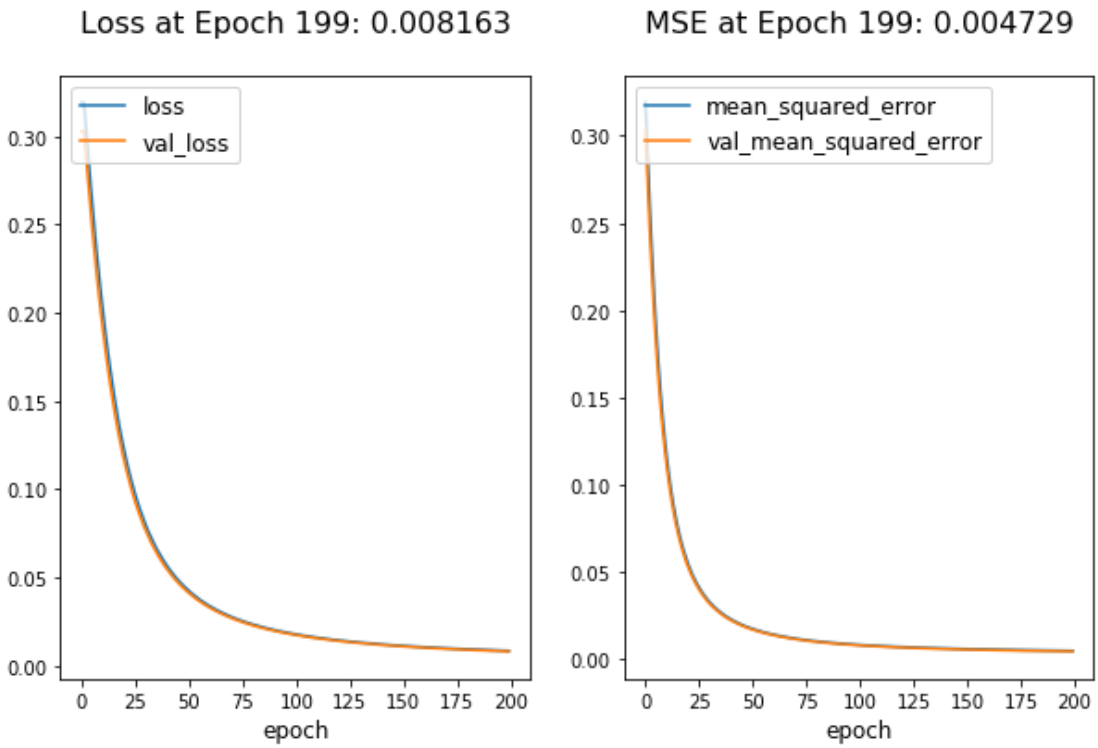

```
> training time is 1.2661 minutes
Train id: 6
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```



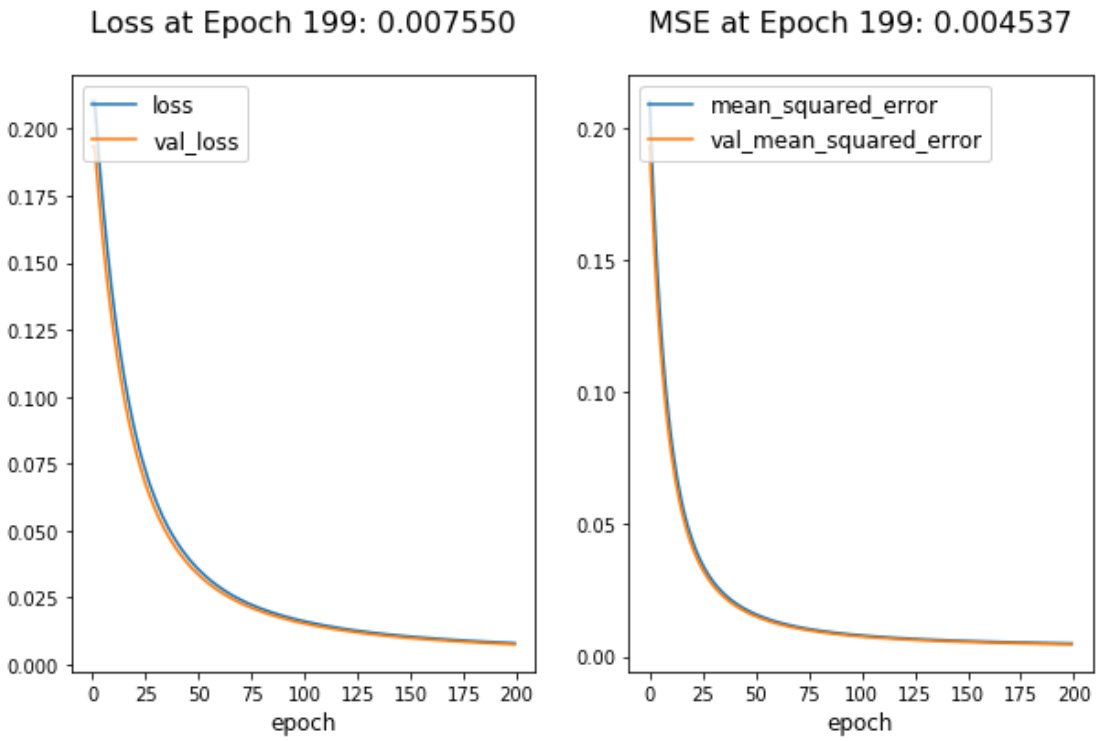

```
> training time is 1.2062 minutes
Train id: 7
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```




```
> training time is 1.1392 minutes
Train id: 8
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```




```
> training time is 1.2439 minutes
Train id: 9
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```




```
> training time is 1.2339 minutes
```

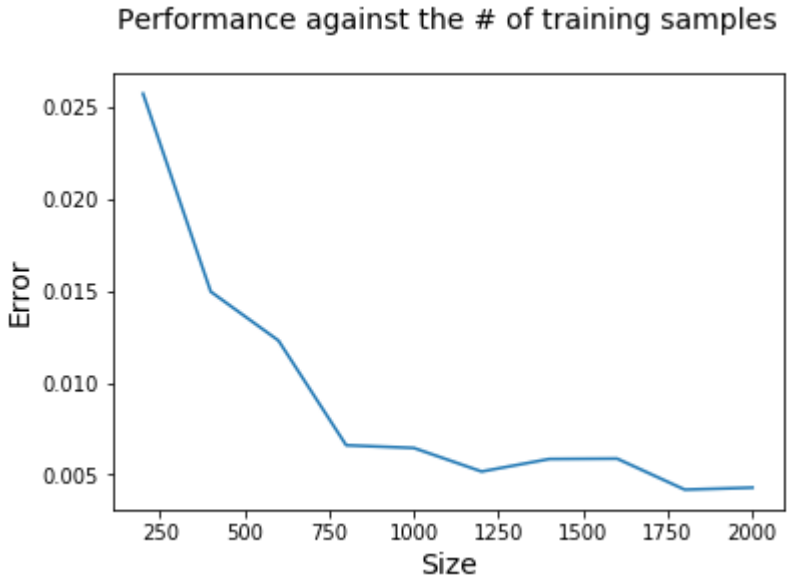
4.4. Evaluating

In [166]:
```
errors = []
for i in range(len(history)):
    epoch_opt = np.argmin(history[i].history["val_mean_squared_erro
r"])
    error = history[i].history["val_mean_squared_error"][epoch_opt]
    errors.append(error)
# for
sizes = list(range(200, 2001, 200))
```

In [183]:
```
vt_min = np.argmin(np.array(errors))
print(f"Min Error at size={sizes[vt_min]} with error = {errors[vt_m
in]}")
```

```
Min Error at size=1800 with error = 0.00419238768517971
```

In [179]:
```
plt.plot(sizes, errors)
plt.xlabel("Size", fontsize = 14)
plt.ylabel("Error", fontsize = 14)
plt.title("Performance against the # of training samples\n", fontsi
ze = 14);
```



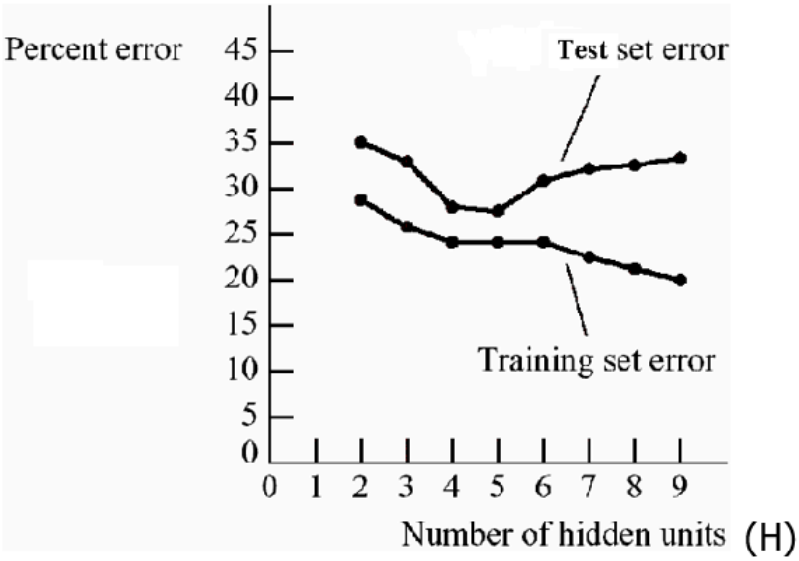
4.5. Discussion

- Performance enhancing with increasing the sample sizes
- The maximize at size = 1800 with error = .00419238768517971

5. EXPERIMENT 3:

In [216]:
```
image = plt.imread("img/exp3.png")
plt.figure(figsize=(14,14))
plt.axis("off"), plt.imshow(image);
```

- Performance against the # hidden nodes (H)
  - Set E=E*, N=N*



- Setup:
  - E* = 200
  - H = 1 - 10
  - N = 1800

5.1. Train Data

In [184]:
```
x_exp3_train, y_exp3_train = x_train[:1800], y_train[:1800]
x_exp3_valid, y_exp3_valid = x_valid, y_valid

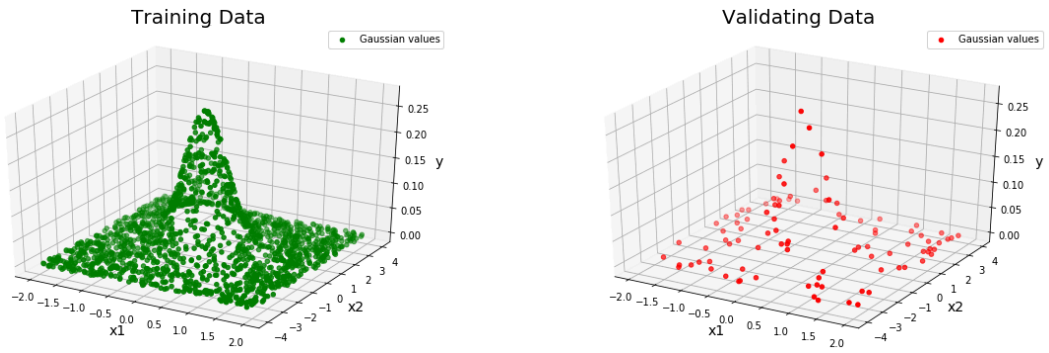
print(f"Train: x-shape = {x_exp3_train.shape} y-shape = {y_exp3_train.shape}")
print(f"Valid: x-shape = {x_exp3_valid.shape} y-shape = {y_exp3_valid.shape}")

plt.figure(figsize=(20, 6))

ax1 = plt.subplot(1,2,1, projection='3d')
visualize_dataset(x_exp3_train, y_exp3_train, color = "green", title = "Training Data\n", ax = ax1)

ax2 = plt.subplot(1,2,2, projection='3d')
visualize_dataset(x_exp3_valid, y_exp3_valid, color = "red", title = "Validating Data\n", ax = ax2)
```

```
Train: x-shape = (1800, 2) y-shape = (1800,)
Valid: x-shape = (100, 2) y-shape = (100,)
```



5.2. Model

In [185]:
```
exp3_model = build_model(H=10)
```

```
Model: "model_2"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```

5.3. Training

In [194]:
```
def train_model(train_id, x_local_train, y_local_train, x_local_val
id, y_local_valid, n_hidden = 10, nb_epochs = 1000):

    K.clear_session()
    exp3_model = build_model(H=n_hidden)

    # OPTIMIZATION
    exp3_model.compile(loss='mean_squared_error', optimizer=SGD(lr=
0.01), metrics=['mean_squared_error'])

    # CALLBACKS
    callbacks = []

    # + ModelCheckPoint
    save_dir = f"./data/model/train_03/{train_id}"
    if os.path.exists(save_dir)==False: os.makedirs(save_dir)
    template_name= "funcApxm_model_01_{epoch:02d}_{val_loss:.3f}_va
l_acc-{val_mean_squared_error:.3f}.h5"
    if os.path.exists(save_dir) == False: os.makedirs(save_dir)
    modekcheckpoint_callback = ModelCheckpoint(filepath = os.path.j
oin(save_dir, template_name),
                                               monitor='val_loss',
                                               verbose=0,
                                               save_best_only=True,
                                               save_weights_only=Fa
lse, # only weight,
                                               mode='auto', # min,
max, auto
                                               period=1  # bao nhie
u epoch moi xem xet 1 lan
                                              )
    callbacks.append(modekcheckpoint_callback)

    # + Early Stopping
    # earlystopping_callback = EarlyStopping(monitor='val_loss', pa
tience= 10)
    # callbacks.append(earlystopping_callback)

    # + Visualize logs
    review_logs = {}
    def custom_on_epoch_end(self, epoch, logs, **kwargs):
        review_logs = kwargs["review_logs"]
        if epoch == 0:
            review_logs["epoch"] = [epoch]
            for key in logs.keys(): review_logs[key] = [logs[key]]
        # if
        review_logs["epoch"].append(epoch)
        for key in logs.keys(): review_logs[key].append(logs[key])

        visualize_data = review_logs
        group_view     = [{"x": "epoch", "y": ["loss", "val_loss"],
                           "title": "Loss at Epoch {epoch}: {val_lo
ss:.6f}\n",
                           "style": "median"},
                          {"x": "epoch", "y": ["mean_squared_erro
```

```
r", "val_mean_squared_error"],
                                "title": "MSE at Epoch {epoch}: {val_mea
n_squared_error:.6f}\n",
                                "style": None}
                               ]
        fig = visualize_logs(visualize_data, group_view, figsize =
(10, 6), is_show = False,
                                 save_path = os.path.join(f"./data/mode
l/train_03/{train_id}", "logs.png"))
        plt.close()
        if epoch == 0:
            display.display(fig, display_id=f"logs_{train_id}")
        else:
            display.update_display(fig, display_id=f"logs_{train_i
d}")
        pass
    # custom_on_epoch_end

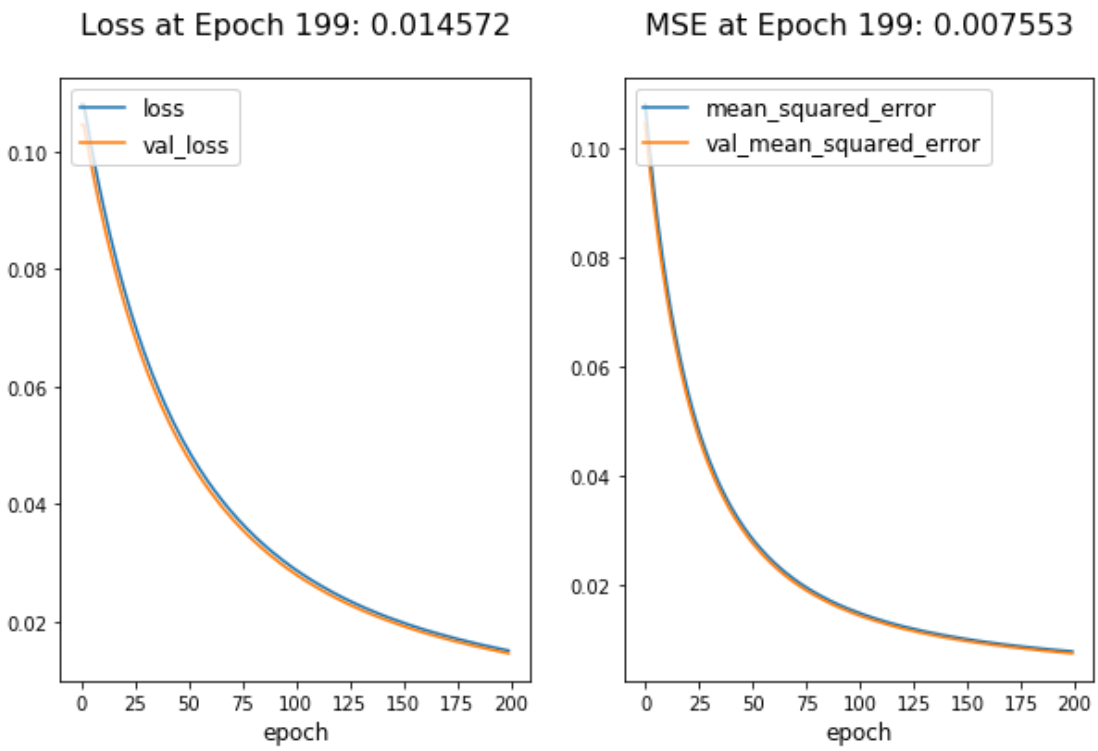
    custom_callbacks = LambdaCallback()
    custom_callbacks.on_epoch_end = lambda epoch, logs: custom_on_e
poch_end(custom_callbacks, epoch, logs, review_logs = review_logs)
    callbacks.append(custom_callbacks)

    # TRAINING
    batch_size = 128
    epochs = nb_epochs

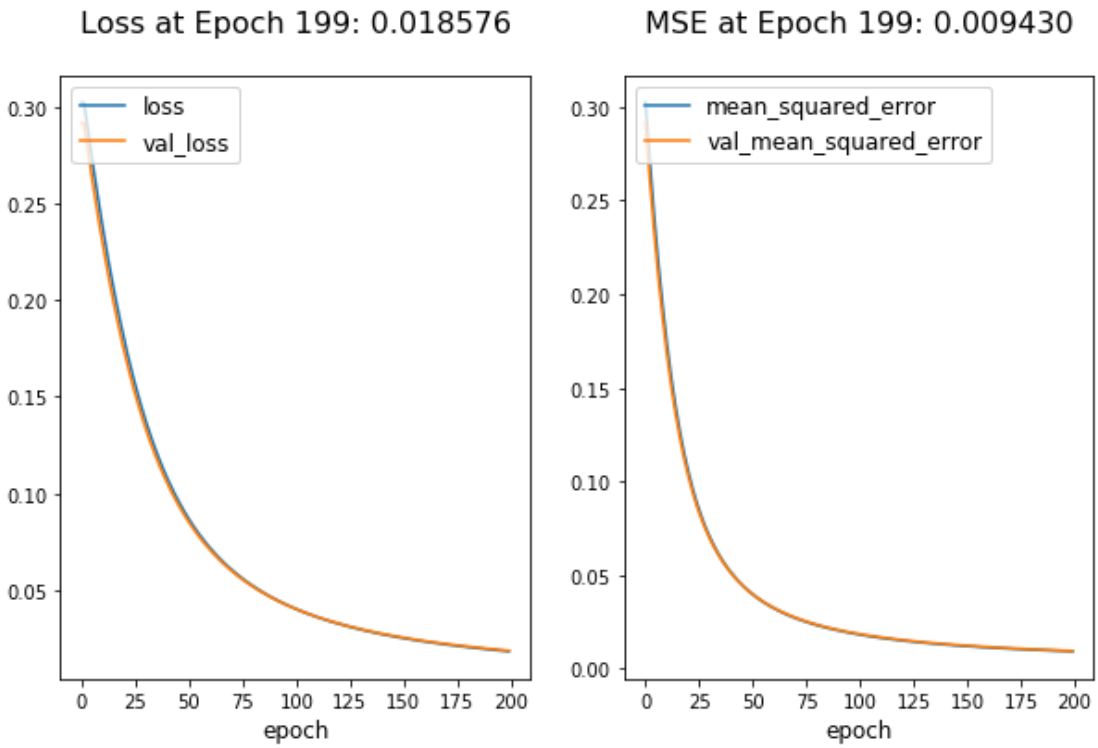
    starting_time = time.time()
    history = exp3_model.fit(x_local_train, y_local_train,
                         validation_data=(x_local_valid, y_local_val
id),
                         batch_size=batch_size,
                         epochs=epochs,
                         callbacks = callbacks,
                         verbose = 0)
    print('> training time is %.4f minutes' % ((time.time() - start
ing_time)/60))
    return history
# train_model
```

In [195]:
```
history = []
for i in range(1, 11, 1):
    print(f"Train id: {i}")
    temp_history = train_model(i, x_exp3_train, y_exp3_train, x_exp
3_valid, y_exp3_valid, n_hidden=i, nb_epochs = 200)
    history.append(temp_history)
# for
```

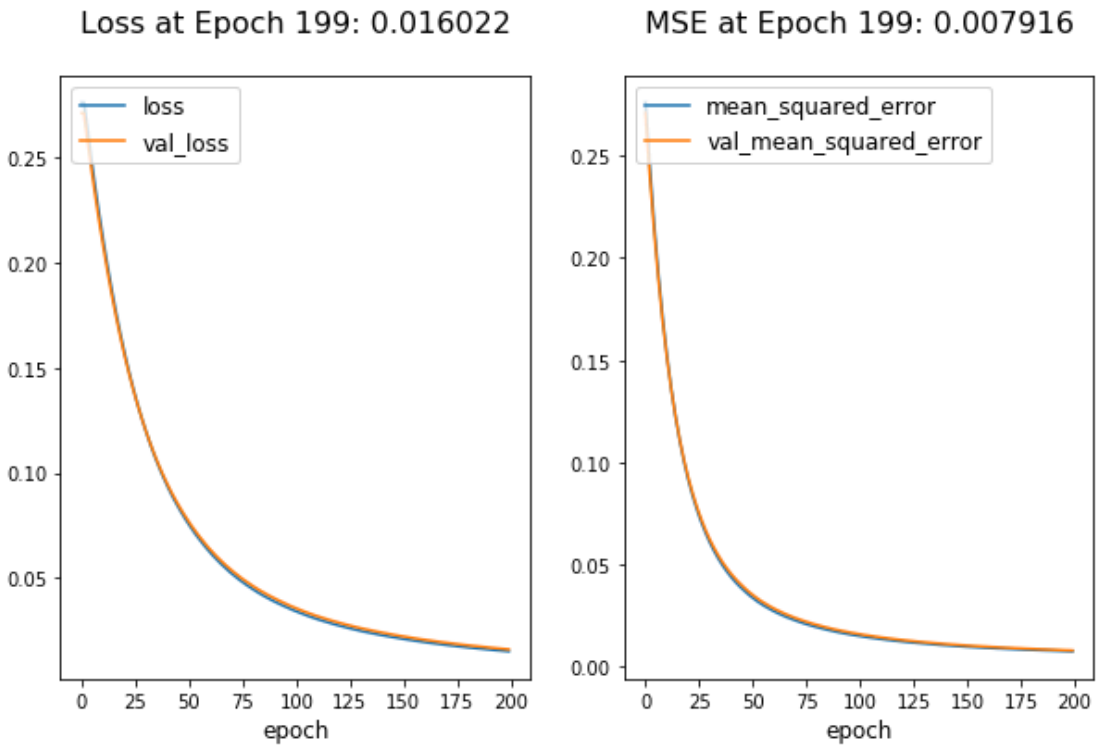
```
Train id: 1
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 1)                 3
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 2
=================================================================
Total params: 5
Trainable params: 5
Non-trainable params: 0
_________________________________________________________________
```



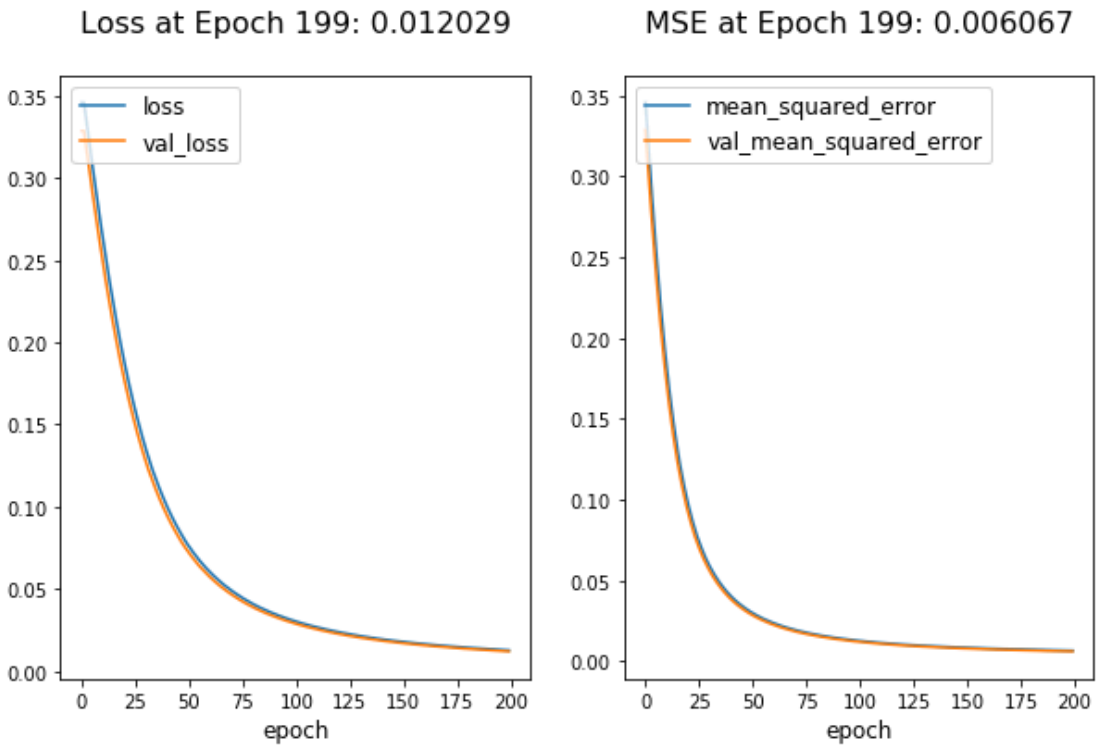

```
> training time is 1.5786 minutes
Train id: 2
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 2)                 6
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 3
=================================================================
Total params: 9
Trainable params: 9
Non-trainable params: 0
_________________________________________________________________
```



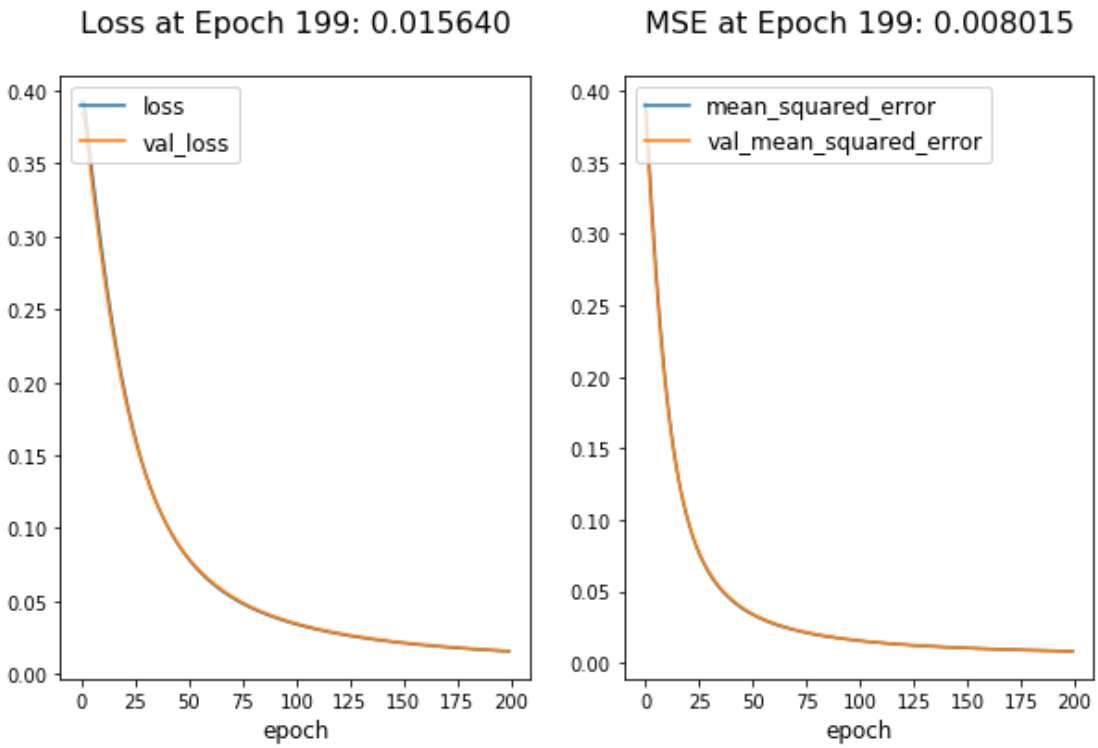

```
> training time is 1.3672 minutes
Train id: 3
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 3)                 9
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 4
=================================================================
Total params: 13
Trainable params: 13
Non-trainable params: 0
_________________________________________________________________
```



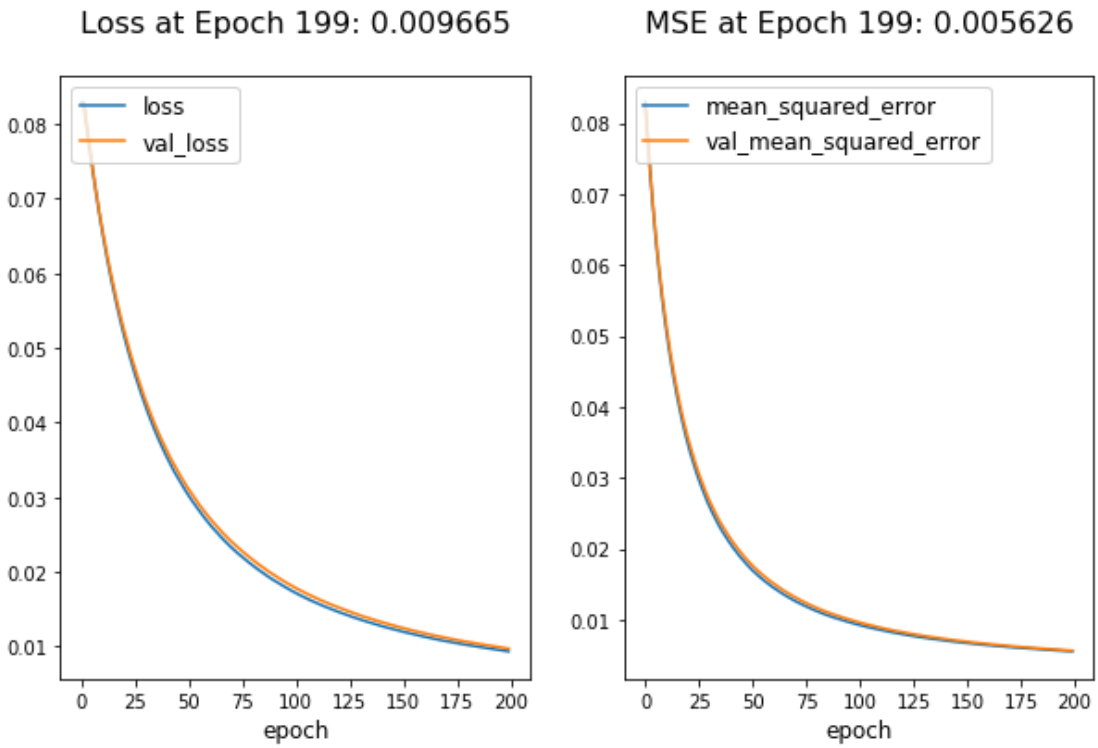

```
> training time is 1.3949 minutes
Train id: 4
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 4)                 12
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 5
=================================================================
Total params: 17
Trainable params: 17
Non-trainable params: 0
_________________________________________________________________
```



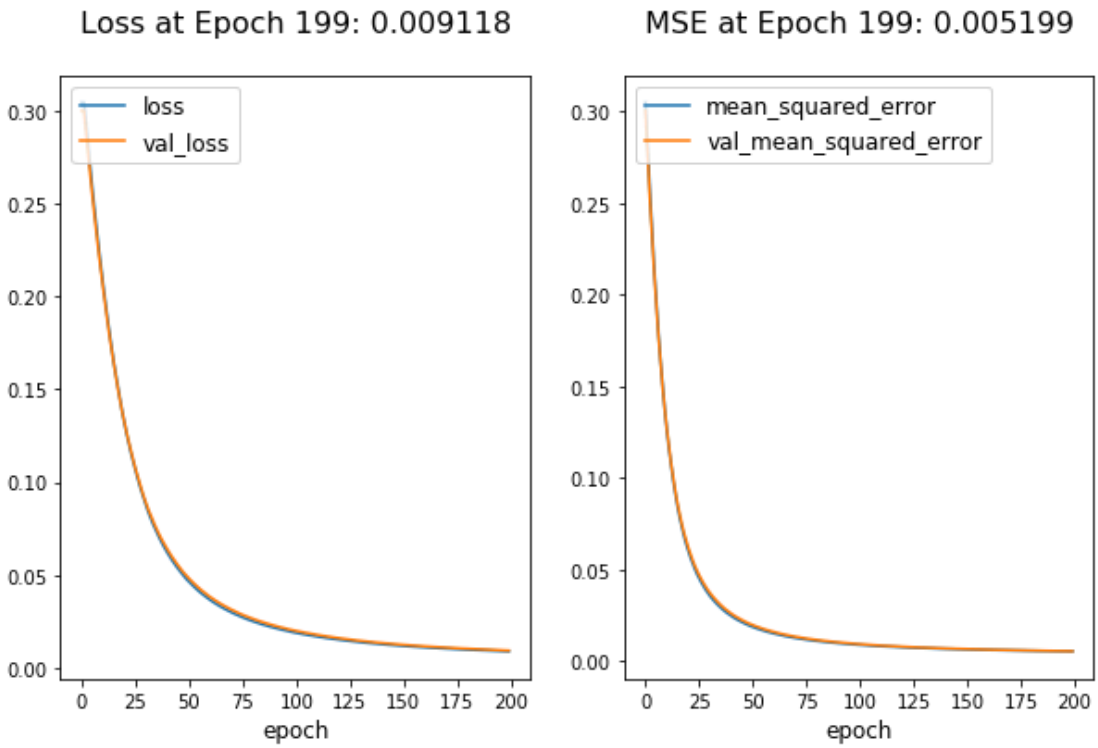

```
> training time is 1.4472 minutes
Train id: 5
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 5)                 15
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 6
=================================================================
Total params: 21
Trainable params: 21
Non-trainable params: 0
_________________________________________________________________
```



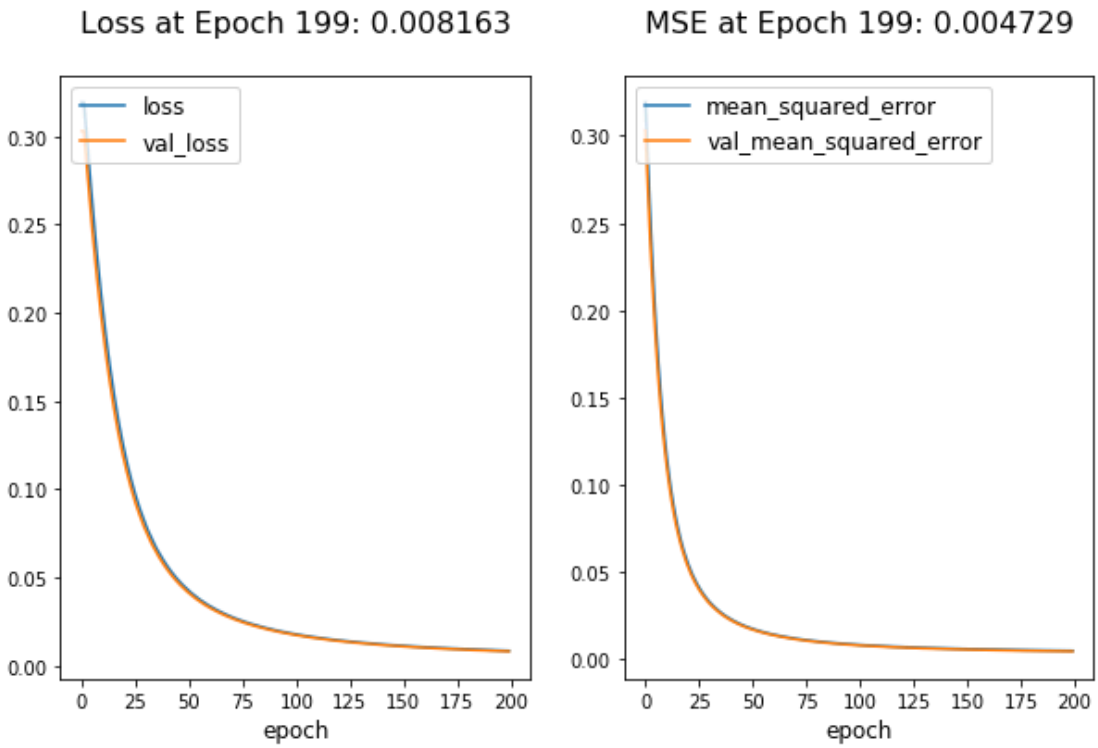

```
> training time is 1.3089 minutes
Train id: 6
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 6)                 18
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 7
=================================================================
Total params: 25
Trainable params: 25
Non-trainable params: 0
_________________________________________________________________
```



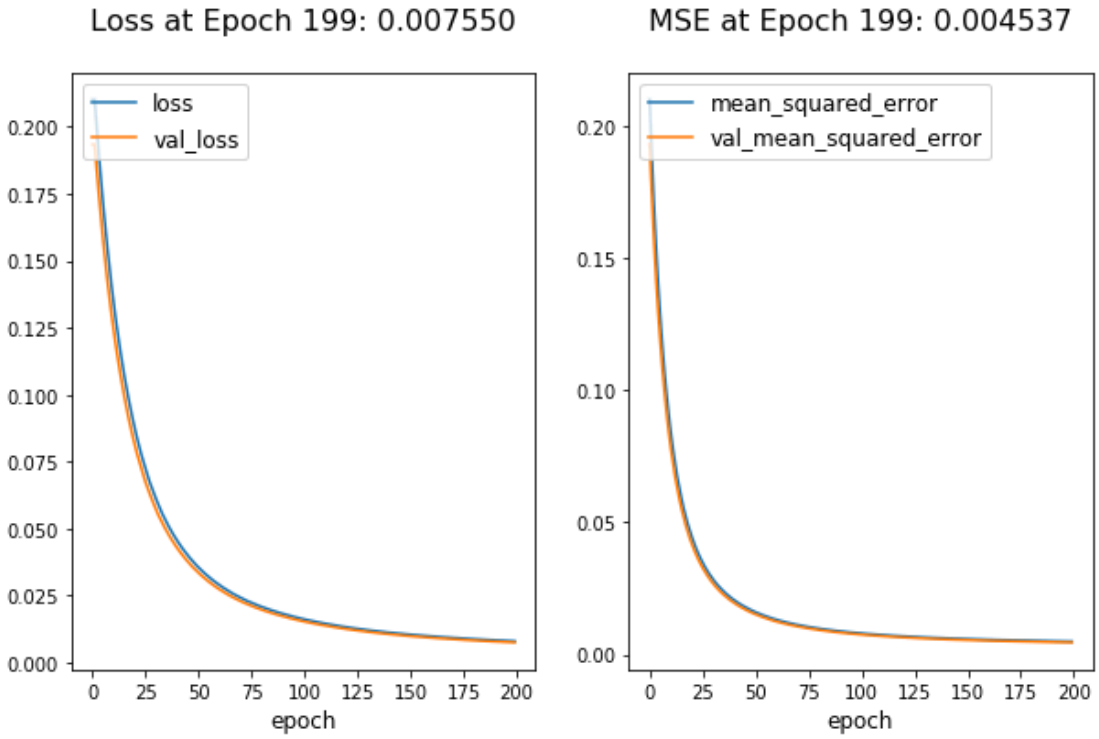

```
> training time is 1.3716 minutes
Train id: 7
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 7)                 21
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 8
=================================================================
Total params: 29
Trainable params: 29
Non-trainable params: 0
_________________________________________________________________
```



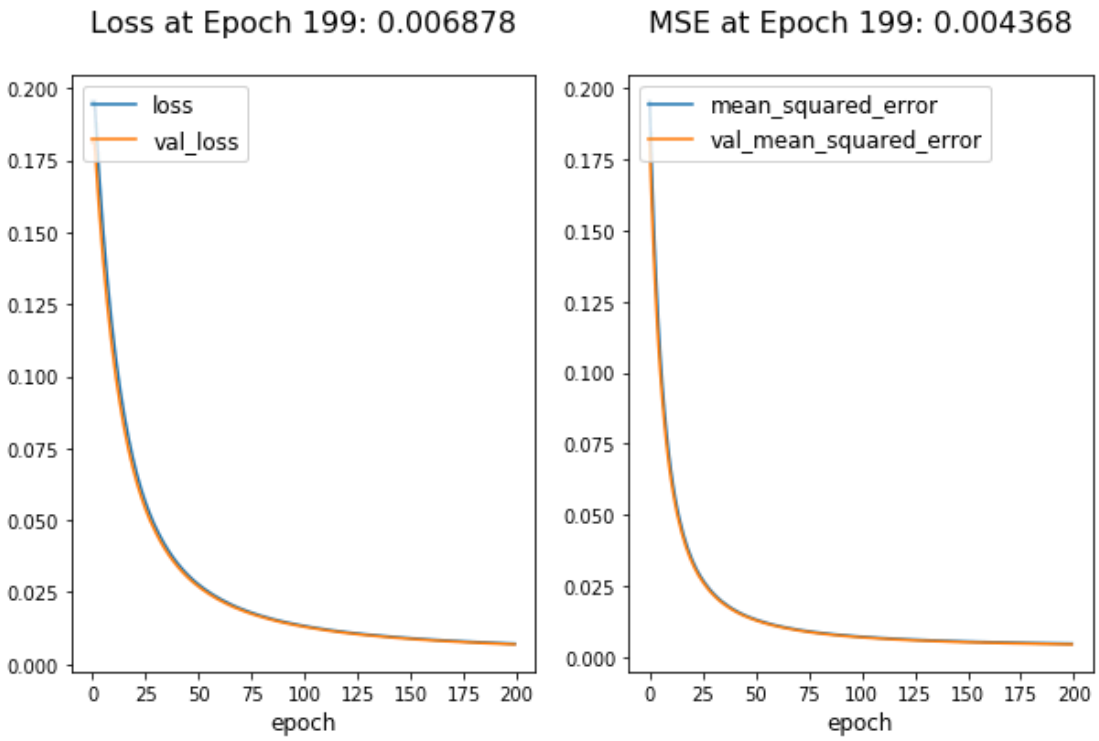

```
> training time is 1.4423 minutes
Train id: 8
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 8)                 24
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 9
=================================================================
Total params: 33
Trainable params: 33
Non-trainable params: 0
_________________________________________________________________
```




```
> training time is 1.4462 minutes
Train id: 9
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 9)                 27
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 10
=================================================================
Total params: 37
Trainable params: 37
Non-trainable params: 0
_________________________________________________________________
```




```
> training time is 1.3184 minutes
Train id: 10
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
Input (InputLayer)           (None, 2)                 0
_________________________________________________________________
fc (Dense)                   (None, 10)                30
_________________________________________________________________
gauss_apxm (Dense)           (None, 1)                 11
=================================================================
Total params: 41
Trainable params: 41
Non-trainable params: 0
_________________________________________________________________
```




```
> training time is 1.3599 minutes
```

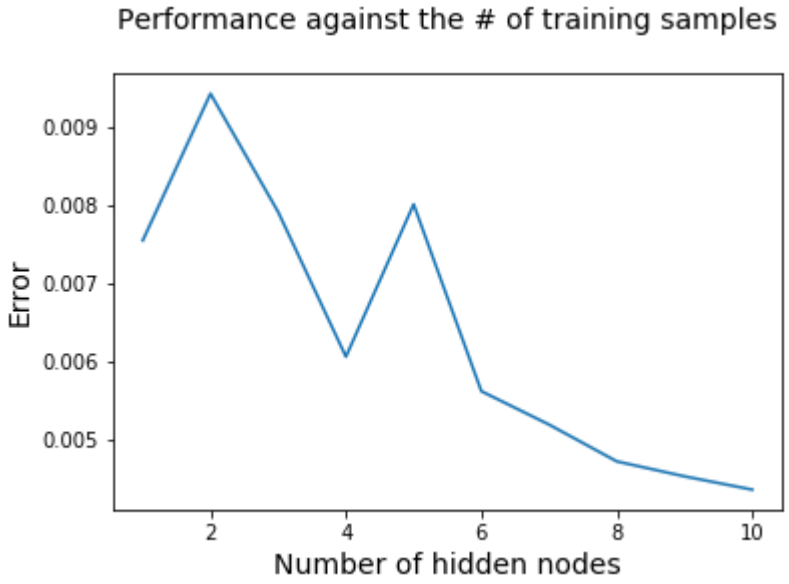
5.4. Evaluating

In [204]:
```
errors = []
for i in range(len(history)):
    epoch_opt = np.argmin(history[i].history["val_mean_squared_erro
r"])
    error = history[i].history["val_mean_squared_error"][epoch_opt]
    errors.append(error)
# for
hiddens = list(range(1, 11, 1))
```

In [205]:
```
vt_min = np.argmin(np.array(errors))
print(f"Min Error at n_hidden={hiddens[vt_min]} with error = {error
s[vt_min]}")
```

```
Min Error at n_hidden=10 with error = 0.004367605317384005
```

In [206]:
```
plt.plot(hiddens, errors)
plt.xlabel("Number of hidden nodes", fontsize = 14)
plt.ylabel("Error", fontsize = 14)
plt.title("Performance against the # of training samples\n", fontsi
ze = 14);
```



5.5. Discussion

- Performance enhancing with increasing the hidden nodes
- The maximize with number of hidden nodes 10 with error = 0.004367605317384005